

Customer Behavior & Shopping Trends Analytics

End-to-End Data Analytics Project Report | Python, PostgreSQL & Power BI

1. Executive Summary

This comprehensive data analytics project investigates consumer shopping patterns, demographic preferences, and product performance using a structured retail dataset. By leveraging an end-to-end analytics stack—comprising **Python** for data ingestion and preprocessing, **PostgreSQL 18** for relational database management and complex exploratory querying, and **Power BI Desktop** for dynamic visualization—this report delivers actionable insights to optimize retail strategy, enhance customer segmentation, and drive revenue growth.

2. Business Problem Statement & Objectives

Modern retail organizations face challenges in understanding nuanced consumer purchasing behavior across diverse demographics, payment methods, and subscription tiers. Without data-driven visibility, marketing and inventory teams risk misallocating capital.

Key Project Objectives:

- Identify primary revenue-generating demographic cohorts and product categories.
- Evaluate the impact of membership tiers and subscription status on purchase frequency.
- Analyze payment method preferences and shipping type efficiencies.
- Build an interactive Executive Dashboard to monitor core Key Performance Indicators (KPIs).

3. Technical Architecture & Tech Stack

The solution architecture follows an industry-standard ETL and analytical workflow:

Pipeline Stage	Tool / Technology	Description
Data Ingestion & Cleaning	Python (Pandas, Jupyter)	Dataset validation, null-check verification, and initial structuring.
Relational Storage & Querying	PostgreSQL 18 / pgAdmin 4	Database creation (`customer_behavior`), advanced SQL aggregation & filtering.
Data Visualization & BI	Power BI Desktop	Interactive dashboard creation with custom cards, bar charts, and donut charts.

4. Key SQL Analytical Findings

Complex SQL queries executed against the PostgreSQL database yielded critical business insights regarding customer revenue contributions:

- **Age Group Revenue Contribution:** Analysis indicates that *Young Adults* and *Middle-aged* segments contribute the highest aggregate purchase amounts, guiding targeted marketing campaigns.
- **Subscription Analysis:** Subscribers demonstrate a higher frequency of previous purchases (*previous_purchases > 5*), validating the ROI of loyalty programs.

- **Category Performance:** Clothing and Accessories emerge as top-performing revenue drivers across multiple seasons.

5. Power BI Dashboard Architecture

The interactive dashboard features a clean, executive UI layout including:

- **KPI Cards:** Displaying total customers (**3.9K**), average purchase amount (**\$59.76**), and average review rating (**3.75**).
- **Visual Breakdowns:** Donut charts illustrating revenue distribution by shipping type and subscription status, alongside clustered bar charts for regional and categorical performance.

6. Strategic Recommendations

Based on empirical findings from SQL and Power BI models, retail leadership should:

- Incentivize non-subscribers with tiered loyalty discounts to bridge the purchase frequency gap.
- Optimize inventory allocation toward high-margin clothing lines preferred by the 20–40 age demographic.
- Streamline express shipping options to further improve customer satisfaction ratings.